Human Presence and Activity Detection

USING CSI ANALYSIS AND DEEP LEARNING

Review 1

Introduction

Our project investigates the use of Wi-Fi Channel State Information(CSI) for human presence and activity detection with the help of standard 2.4 GHz WiFi router acting as a transmitter and an ESP-32 acting as a receiver.
Human presence and movement induces measurable disturbances in the amplitude and phase of the received Wi-Fi packets, which are captured through CSI measurements. These variations are processed and transformed into sample–Sub carrier frequency spectrogram representations.
By leveraging these spectrograms,we can train deep learning models like EfficientNetV2 for classification of presence and various different human activity.

Wi-Fi Channel State Information (CSI) provides a non-intrusive and privacy-friendly human sensing mechanism by analyzing the changes in the frequency induced by human motion also recent developments in deep learning enable the learning of complex movement patterns from the spectrogram generated by having 400 samples in X axis & 52 subcarrier frequencies in Y axis. (The Color deviation is due to the frequency change).

Motivation

we were motivated by a similar experiment we did during our laser diode diode experiment and then found data can be carried by any kind of waves which motivated us to find various research papers(---)
and develop this project

- To develop a low-cost, scalable, and non-vision-based human detection and activity module.
- To demonstrate how commodity hardware (ESP32, Wi-Fi router, Raspberry Pi) can be used for intelligent sensing.

Objectives

- To design a Wi-Fi CSI-based human presence and activity detection module.
- To design and apply directional antennas for better CSI data collection.
- To apply deep learning models like EfficientNet-V2 for human presence and activity detection by analysing CSI spectrograms.
- To assess the performance of the system for various human presence and activity detection then further improvements by classifying various types of activities.

Hardware

The hardware setup consists of four ESP32-S3-Mini-1U modules, and a TP-Link Router Both Modified to support external antennas and operating in a transmitter–receiver configuration. CSI packets are captured at a fixed sampling rate of 100 Hz, enabling high temporal resolution. The receiver forwards CSI data over a serial interface, where a Python-based data acquisition pipeline records, processes, and visualizes the measurements.

Rather than generating custom packets, the system passively captures CSI from standard Wi-Fi traffic, such as beacon and data frames transmitted by a mobile phone hotspot or the modified router. This approach ensures that the sensing framework remains non-intrusive, infrastructure-light, and compatible with real-world deployments.

Raw CSI measurements often contain impulsive noise and outliers caused by interference, hardware imperfections, and sudden channel fluctuations. To address this, a Hampel filter is applied during post-processing to suppress outliers while preserving meaningful channel dynamics. The filtered CSI streams are then transformed into spectrograms, which serve as robust feature representations for presence detection and activity recognition tasks.

The filtered CSI data is transformed into time–frequency spectrograms, which serve as compact and discriminative feature representations the Spectrogram Characteristics consists of Time (CSI sample index) in x- axis is about 400 samples and Y axis consists of Subcarrier index (frequency bins) that are 52 Wi Fi OFDM data subcarriers. The Pixel intensity: Normalized CSI magnitude also the Image resolution is 52 × 400 pixels. Each spectrogram frame is generated directly from CSI magnitude, without interpolation or resizing, ensuring a one to one correspondence between CSI samples and image pixels.

- **ESP-32 S3 MINI-1U**
  A tiny Wi-Fi + Bluetooth LE module with a dual-core microcontroller.
- **Raspberry pi 4b+ & Display**
  A credit-card sized single-board computer that can run Linux, act as a desktop PC, media center, or embedded controller.
- **Biquad Antenna**
  A directional wire antenna made from two square loops (bi-quad = two squares) that gives higher gain and focused signal compared to a simple dipole.
- **TP-Link Wifi Router**
  Dual-band (2.4 GHz + 5 GHz) Wi-Fi with Gigabit Ethernet and MU-MIMO for better performance with multiple devices.
- **Passive Components**
  Capacators,Resistors,data cables..

Software

ESP-IDF is used for flashing the C++ code which is used for CSI data extraction. This involves handling and processing the CSI matrix to produce OFDM data.
Python based pipeline which Parses CSI output then Handles variable CSI lengths Applies filtering and normalization then generates spectrogram images continuously which is then saved sequentially for downstream processing and learning tasks

RWe trained presence detection and activity recognition models using the standard torchvision.models implementation of the EfficientNetV2 small architecture [15]. Presence detection models, namely PLOS and PNLOS, are trained on DPLOS and DPNLOS, respectively. Additionally, a presence detection model for both scenarios, PLOSNLOS, is trained on DPLOS ∪ DPNLOS. The same approach is used for activity recognition, resulting in models ALOS, ANLOS, and ALOSNLOS.

The same approach is used for activity recognition, resulting in models ALOS, ANLOS, and ALOSNLOS. All models are trained from scratch and undergo 1k epochs of training using the Adam optimizer with a learning rate of 0.0001 and a batch size of 16. To eliminate class imbalances in the training dataset, a balanced sampler is employed. Data augmentation includes random circular shifts along the time axis applied to the spectrograms.

For each model, ten independent training runs are conducted, and the mean and standard deviation of metrics such as precision, recall, F1-score, and classification accuracy are reported based on the respective test datasets.

Python

- Libraries:PyTorch,Numpy,os and sys.

Raspbian OS

ESP IDF used for CSI extraction

EfficientNet-V2 (CNN Model)

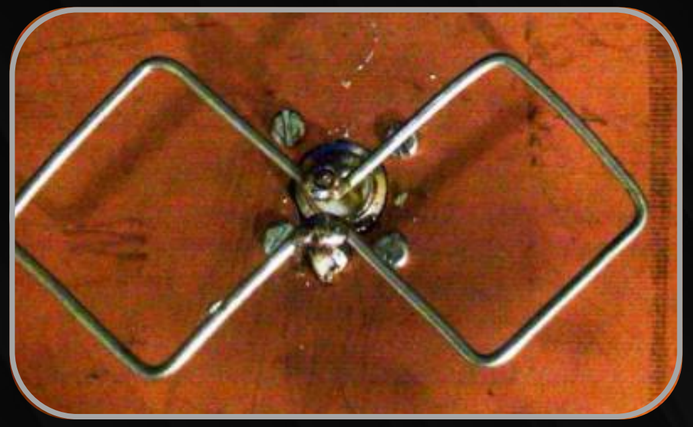
- Description:

  EfficientNetV2-Small (often denoted as EfficientNetV2-S) is a state-of-the-art AI model for computer vision, designed by Google to be smaller and train up to 11 times faster than previous models while maintaining high accuracy. It is used for tasks like identifying objects in images (classification)

Key Components









Applicability

This project will highlight the potential of fetching data using Wi-Fi signals from commonly available devices and processing them through microcontroller-based devices to collect CSI data and use deep learning techniques to detect human presence and activity.



Potential Application Regions

- Emergency and Disaster Response especially detection of people trapped in debris and rubble.
- Defence and Security Reconnaissance
- Non-Invasive Smart Home Security System



A non-intrusive, privacy-preserving sensing approach applicable across safety, security, and smart environments.

Inspiration

- **Citations and References:**

- https://developer.espressif.com/blog/getting-started-with-wifi-on-esp-idf/
- https://github.com/StevenMHernandez/ESP32-CSI-Tool.git
- https://github.com/ashus3868/Wi-Fi-Motion-Detector-ESP32.git
- https://www.mathworks.com/help/dsp/ref/hampelfilter.html
- https://github.com/aiotgroup/Person-in-WiFi-3D-repo.git
- https://github.com/StevenMHernandez/ESP32-CSI-Tool.git
- https://www.youtube.com/watch?v=7XwgxTXPJKM
- https://docs.espressif.com/projects/espressif-ide/en/latest/
- https://www.researchgate.net/publication/374056096_WiFi_CSI-Based_Long-Range_Through Wall_Human_Activity_Recognition_with_the_ESP32

Review 2

PLAN OF ACTION

PHASE 1> Design the PCB and antenna and perform initial hardware and antenna testing.

PHASE 2> Integrate hardware with software, train DL models on public datasets, and finalize the system architecture.

PHASE 3> Refine hardware/software based on results

PHASE 1

Hardware Design and CSI Data Acquisition

In Phase 1, the goal is to establish the basic sensing infrastructure required for Wi-Fi based human detection using Channel State Information (CSI). This phase focuses on hardware setup, antenna development, and validation of CSI data extraction.

Component Assembly

Transmitter:

The transmitter consists of a standard 2.4 GHz Wi-Fi router configured to continuously transmit Wi-Fi packets such as beacon frames and data frames. These packets act as the carrier signal that propagates through the environment and interacts with surrounding objects and human motion.

Receiver:

The receiver is built using ESP32-S3-Mini-1U modules configured to capture CSI measurements from the received Wi-Fi packets. The ESP32 modules are programmed using ESP-IDF firmware to extract the CSI matrix associated with each received packet.

Antenna Development

Directional biquad antenna which will operate at 2.4 GHz is designed and fabricated to improve signal reception and reduce multipath interference. The antenna is tested to ensure stable signal gain and proper impedance matching with the ESP32 module.

CSI Data Generation

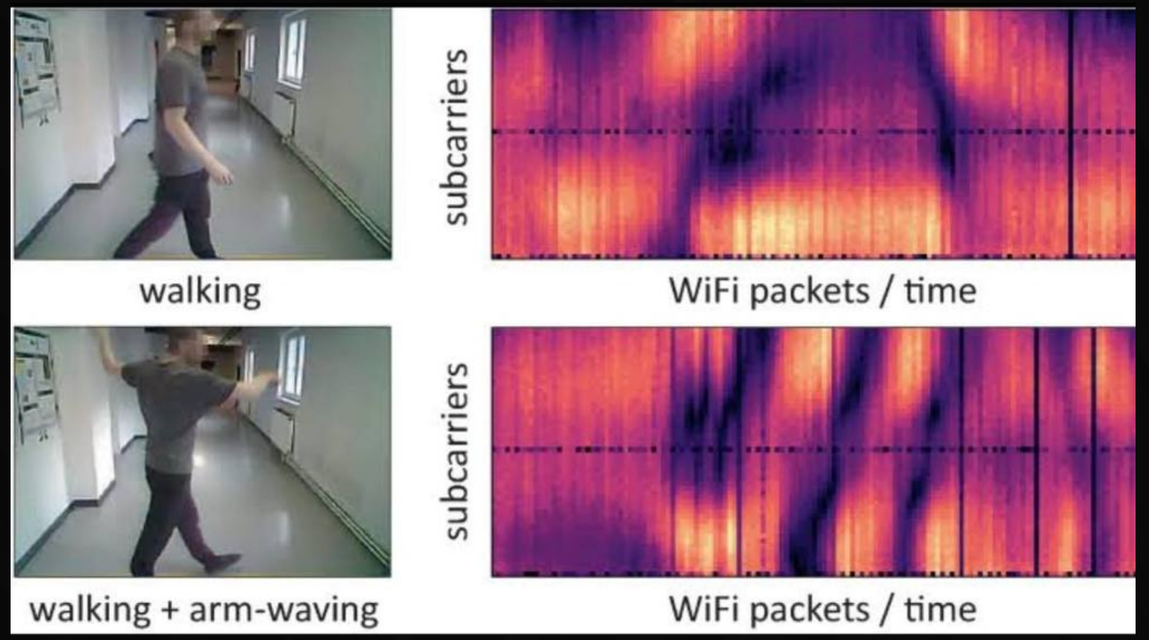
1. Establish a transmitter–receiver configuration between the Wi-Fi router and ESP32 receiver modules.
2. Configure the ESP32 to capture CSI packets at a sampling rate of approximately 40 Hz.
3. Verify the stability of CSI data by monitoring amplitude variations across 52 OFDM(Orthogonal Frequency-Division Multiplexing) subcarriers.
4. Perform initial hardware validation to ensure consistent data acquisition.

PHASE 1

Hardware Design and CSI Data Acquisition

Expected Outcome(Already Achieved)

- Successful fabrication and testing of the custom PCB and antenna system.
- Stable extraction of raw CSI data streams from Wi-Fi packets.
- Validation that human movement introduces measurable disturbances in the CSI signal.



PHASE 2

System Integration and Deep Learning Model Development

CSI Data Processing

1. Data Acquisition Pipeline:
2. Develop a Python-based pipeline to receive CSI data from the ESP32 through a serial interface.
3. Data Cleaning and Filtering:
4. Apply signal processing techniques such as the Hampel filter to remove impulsive noise and outliers from the CSI measurements.
5. Feature Representation:
6. **Transform the filtered CSI signals into time–frequency spectrograms, where:**
   - X-axis represents time samples (~400 samples)
   - Y-axis represents subcarrier indices (52 OFDM subcarriers)
   - Pixel intensity represents normalized CSI magnitude

Deep Learning Model Implementation

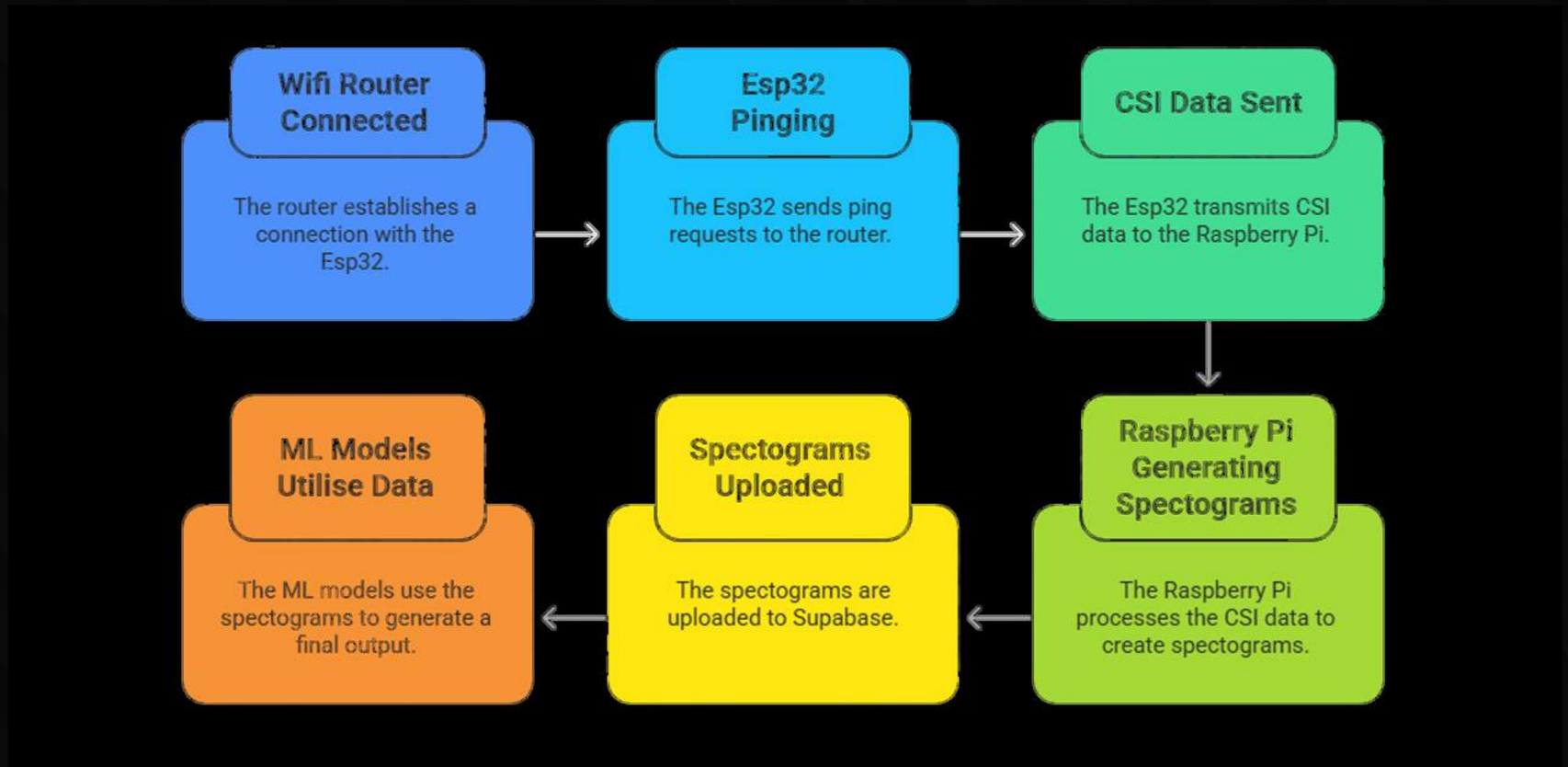
1. Model Selection:
2. Implement EfficientNetV2-Small, a convolutional neural network optimized for image classification.
3. Training Dataset:
4. Use public CSI datasets along with generated spectrogram data to train the model.
5. Training Process:
6. Train the model using the Adam optimizer, a learning rate of 0.0001, and data augmentation techniques such as time-axis circular shifts.
7. Performance Evaluation:
8. **Evaluate model performance using metrics including:**
   - Precision
   - Recall
   - F1-Score
   - Classification Accuracy

PHASE 2

System Integration and Deep Learning Model Development

Expected Outcome(Already Achieved)

- Successful integration of hardware sensing and software processing pipeline.
- Generation of spectrogram representations from CSI data.
- Trained deep learning models capable of detecting human presence and activities.
- Initial performance results demonstrating promising recall and classification accuracy.



PHASE 3

System Optimization and Real-World Deployment

In **Phase 3**, the focus will be on refining the system to improve performance, reliability, and real-world applicability.

Error Analysis and System Improvement

1. Model Optimization:
   Improve classification accuracy by expanding the training dataset and tuning hyperparameters of the deep learning model.
2. Hardware Optimization:
   Optimize antenna placement, transmission power, and receiver positioning to reduce environmental noise and improve CSI stability.

3.Environment Testing:
   Conduct experiments in different scenarios such as:
   - Line-of-Sight (LOS)
   - Non-Line-of-Sight (NLOS)
   - Indoor cluttered environments

4.Real-Time Detection System:
   Develop a real-time inference pipeline capable of detecting human presence and activities continuously.

5.Deployment and Visualization:
   Create a deployable system using Raspberry Pi or cloud infrastructure, along with a visualization dashboard to monitor activity predictions.

PHASE 3

System Integration and Deep Learning Model Development

Expected Outcome(Yet to Achieve)

- Improved robustness and accuracy of the detection system.
- A fully functional real-time Wi-Fi CSI-based human sensing system.
- Demonstration of practical applications such as smart home monitoring, security systems, and disaster rescue scenarios.

VIT-AP

Review 3

Final System Integration

Expected Transition from prototype → working system

All modules integrated successfully

Real-time pipeline established

In Review-3, we focused on complete system integration and real-time deployment.

Complete System Architecture

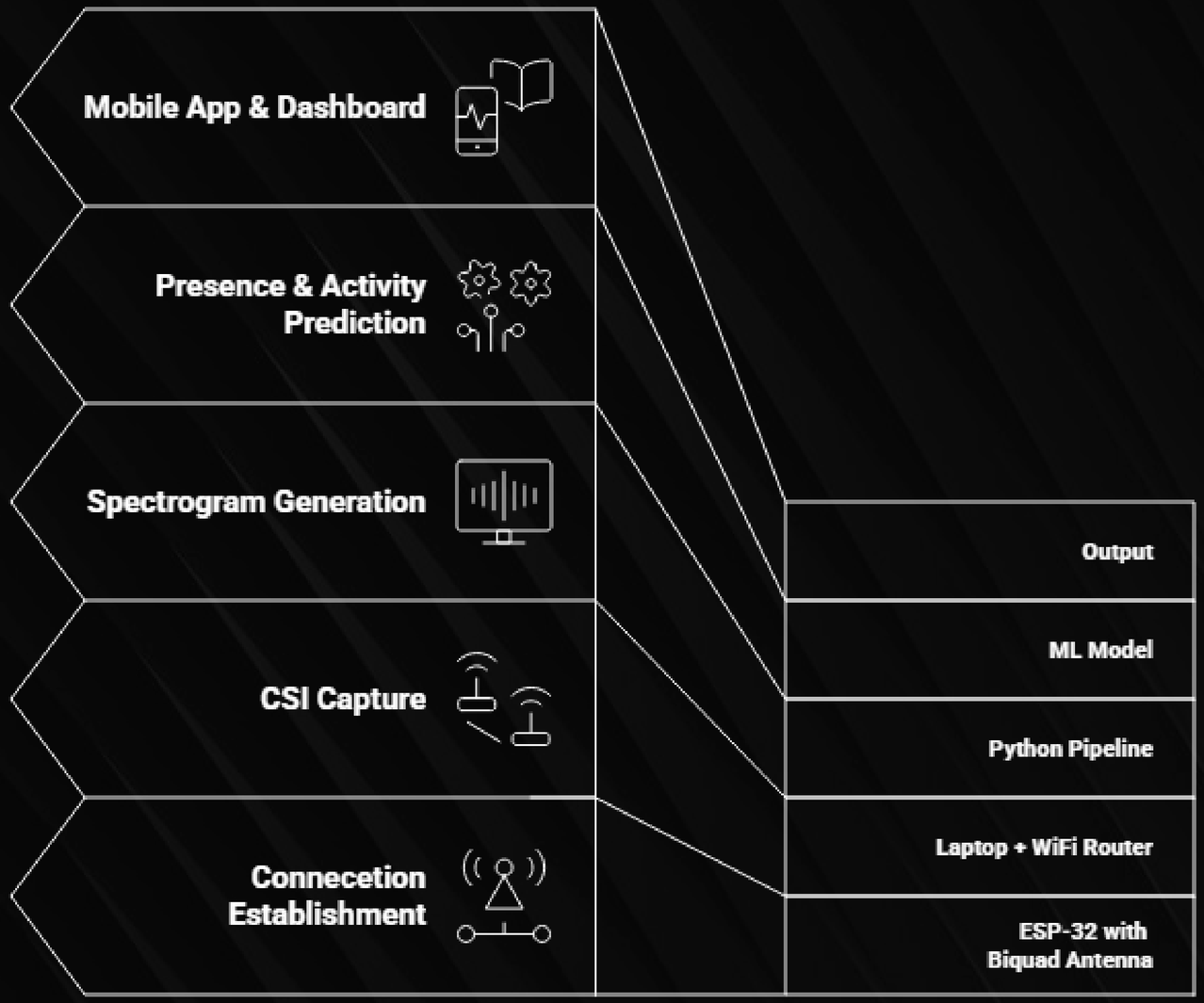
Predictions are displayed on mobile app and dashboard

Deep learning model analyzes spectrograms to predict presence and activity

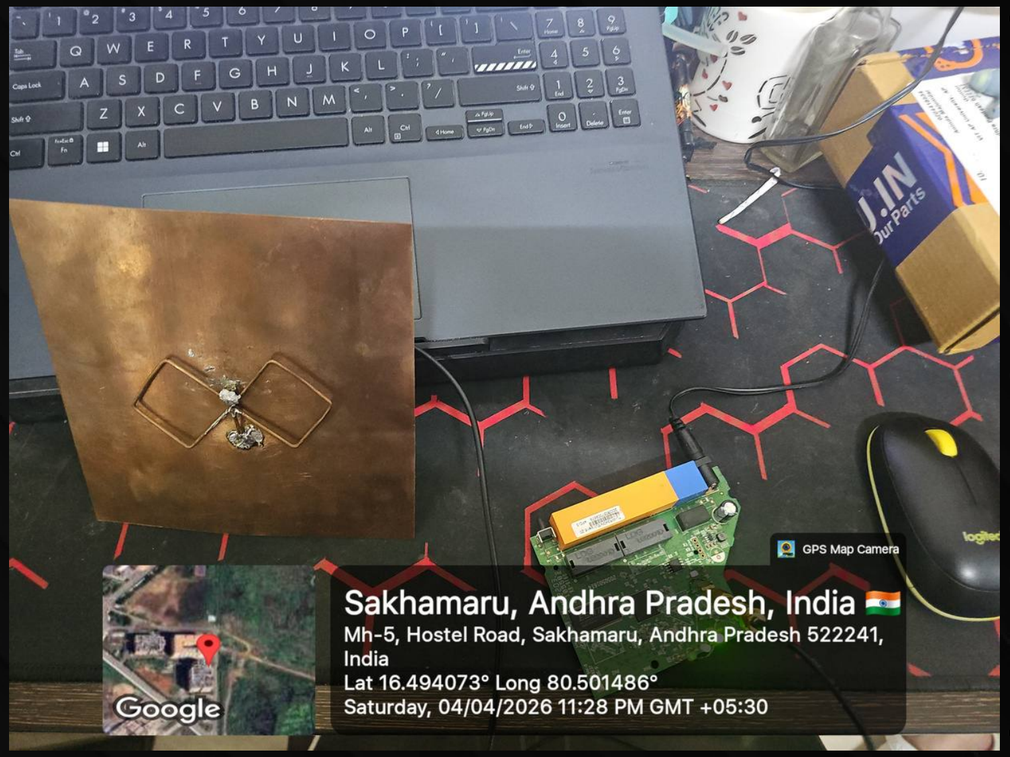
Python script processes CSI Matrix to create spectrograms and send them to databse

ESP32 and laptop exchange pings to establish connection and collect CSI matrix

WiFi router transmits signals captured by ESP32 with Biquad antenna and laptop



Hardware Implementation



Transmitter

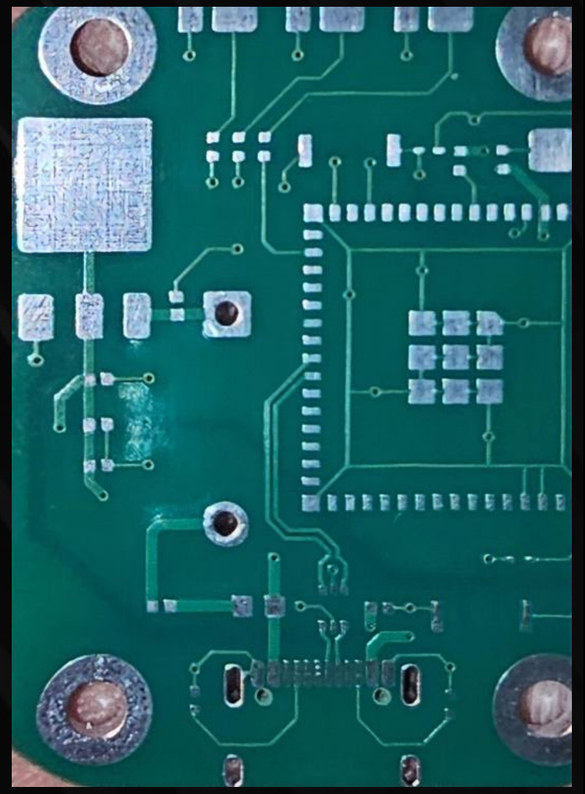


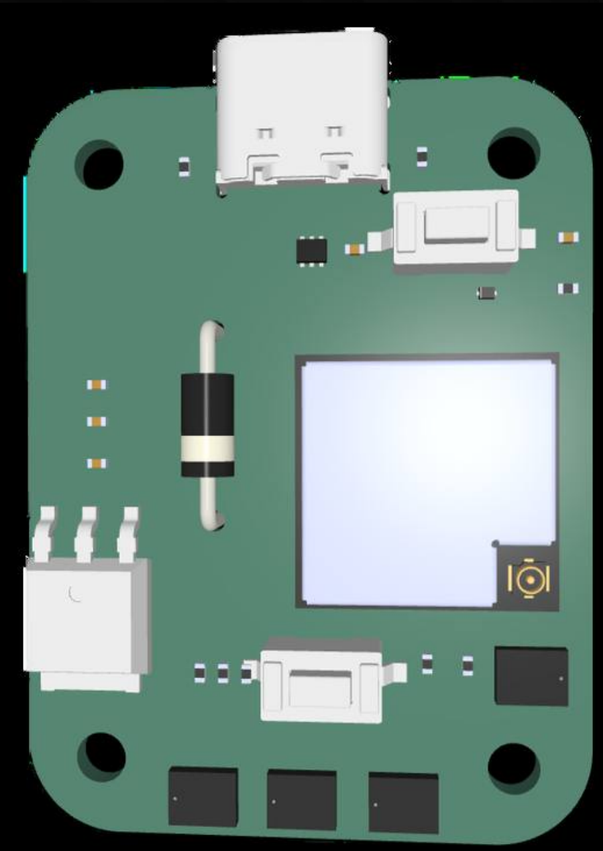
Reciver

Custom PCB & Antenna



Biquad Antenna





Custom PCB

DeepLearning Pipelining

Data Input

CSI spectrogram images (PNG format)

Dataset Handling

Train/Validation/Test splits and class imbalance handling

DataLoader

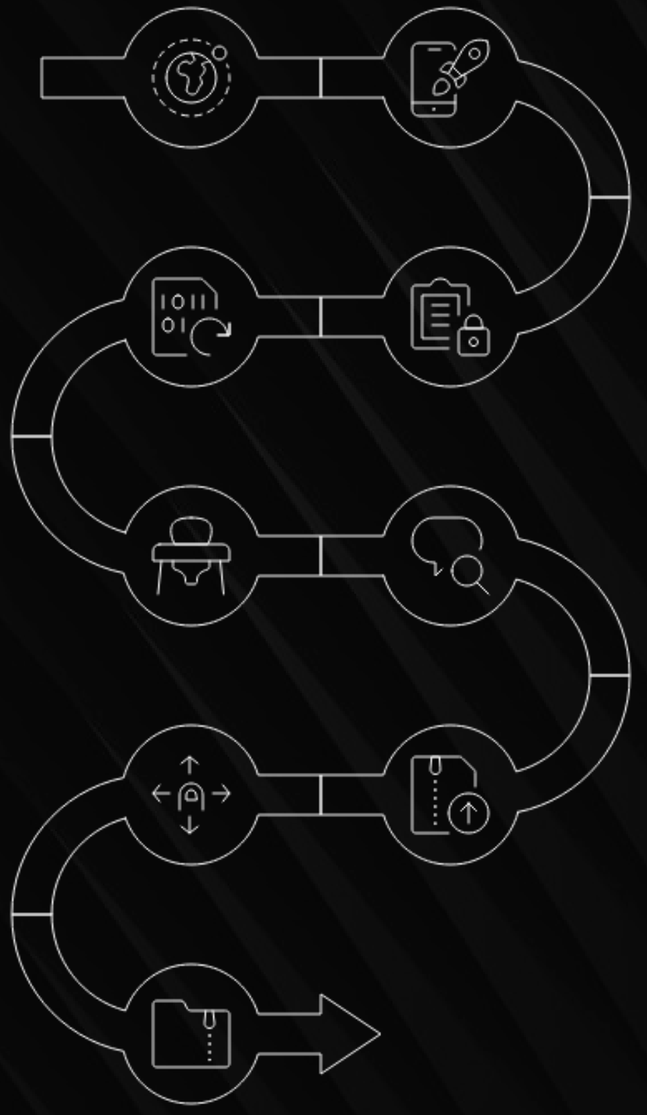
Batch loading and multi-worker loading

Evaluation

Accuracy, Precision, Recall, F1 Score, Confusion Matrix

Inference

Predict class labels and probabilities



Preprocessing

Convert images to tensor and normalize

Data Augmentation (Training Only)

Random Circular Shift and Gaussian Noise

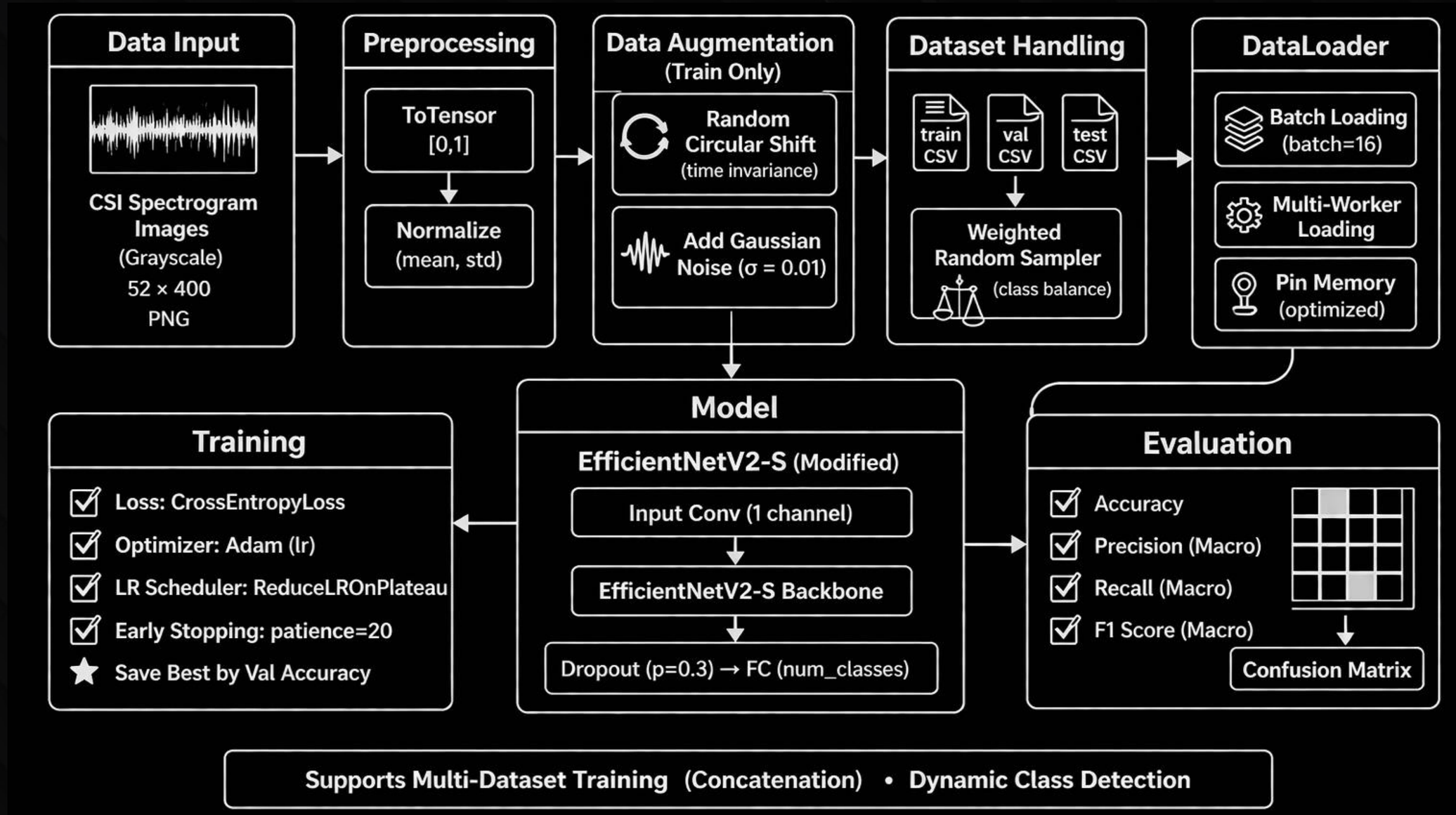
Model

EfficientNetV2-S backbone with modifications

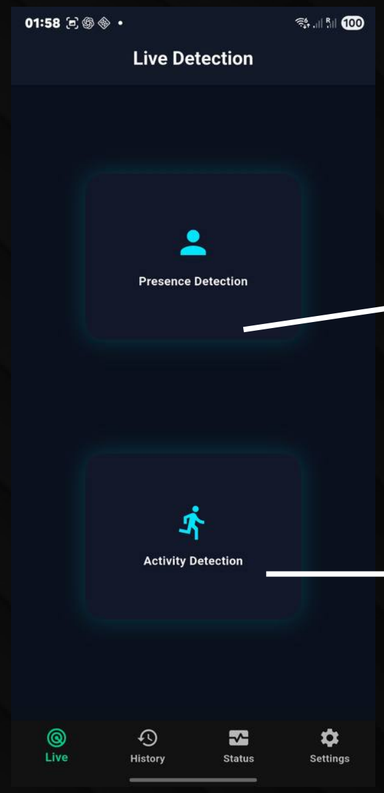
Training

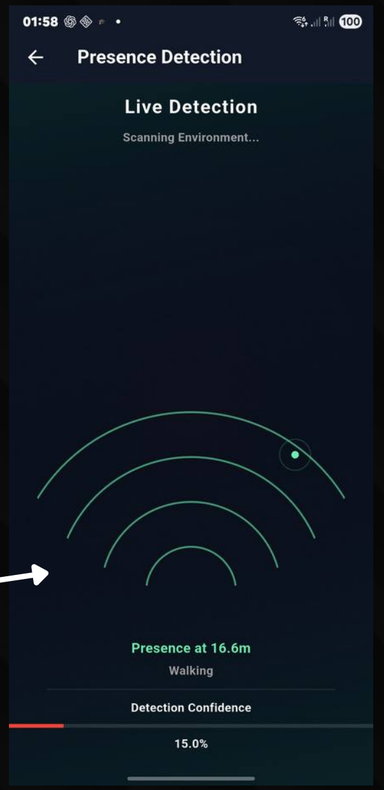
CrossEntropyLoss, Adam optimizer, LR scheduler, Early Stopping

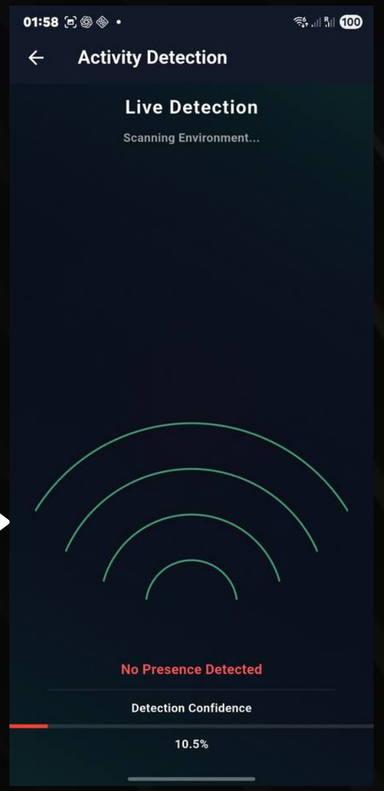
DeepLearning Pipelining

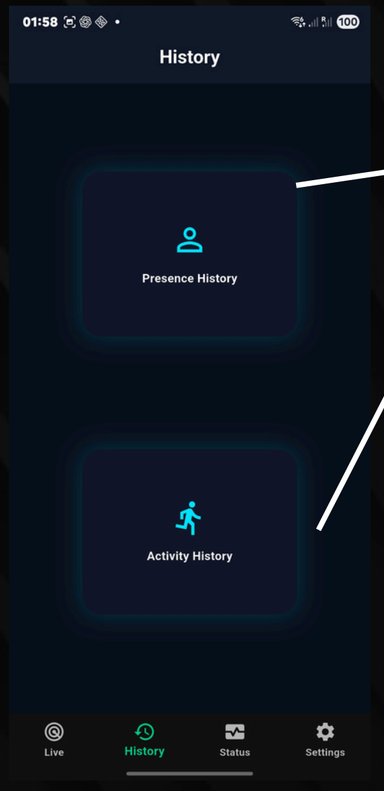


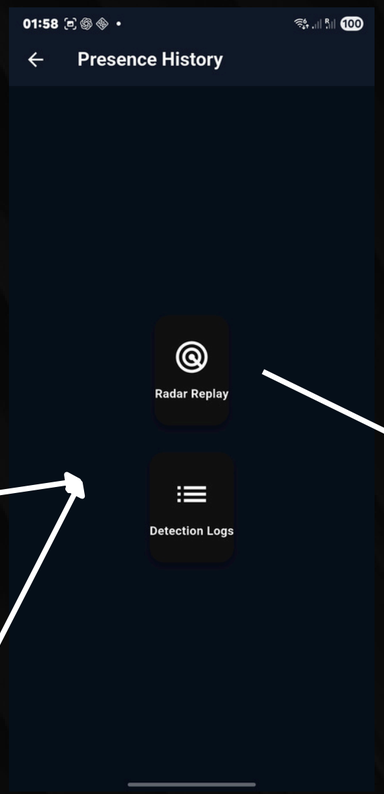
OUTPUT


01:58
Live Detection
Presence Detection
Activity Detection
Live
History
Status
Settings


01:58
Presence Detection
Live Detection
Scanning Environment...
Presence at 16.6m
Walking
Detection Confidence
15.0%


01:58
Activity Detection
Live Detection
Scanning Environment...
No Presence Detected
Detection Confidence
10.5%


01:58
History
Presence History
Activity History
Live
History
Status
Settings


01:58
Presence History
Radar Replay
Detection Logs


01:58
Radar Replay
Radar Replay (presence)
presence event #0
Timestamp
presence event #1
Timestamp
presence event #2
Timestamp
presence event #3
Timestamp

OUTCOME

Objectives

- To design a Wi-Fi CSI-based human presence and activity detection module.
- To design and apply directional antennas for better CSI data collection.
- To apply deep learning models like EfficientNet-V2 for human presence and activity detection by analysing CSI spectrograms.
- To assess the performance of the system for various human presence and activity detection then further improvements by classifying various types of activities.

PHASE 3

System Integration and Deep Learning Model Development

Expected Outcome(Yet to Achieve)

- Improved robustness and accuracy of the detection system.
- A fully functional real-time Wi-Fi CSI-based human sensing system.
- Demonstration of practical applications such as smart home monitoring, security systems, and disaster rescue scenarios.

All proposed outcomes & objective have been successfully achieved and validated through real-time implementation.

Thank You

Hoping for more ideas, inspiration, suggestion and solutions